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Knowledge of AI use and prediction of AI adoption among selected residents in the greater Kumasi area of Ghana

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Abstract

Artificial intelligence (AI) technologies are rapidly changing communities and societies and have had a significant impact on the way of life of individuals. Yet, understanding the influence of AI adoption among people of different sociodemographic background Asocio-demographic, especially in non-Western settings, remains limited. This study examines the knowledge of AI use and the prediction of AI adoption among residents in the Greater Kumasi area of Ghana. The study employed a cross-sectional survey of 407 residents using a multistage sampling technique. The study examined how socio-demographic factors (age, education, gender) influence the perceived usefulness of AI among residents in the Greater Kumasi area and further assessed how AI knowledge and confidence shaped the perceived ease of use of AI among the selected residents. Data was analysed both descriptively presented using frequency distribution and inferentially by way of a binary logistic regression, with a p-value < 0.05 considered statistically significant. Results show that most participants are aware of and use AI with high confidence. The logistic regression analysis showed that the level of education, age, knowledge and confidence in AI are determinants of AI adoption. Our findings reflect the literature on technology adoption pathways. These findings contribute to AI adoption literature in non-Western settings with implications for promoting AI education with targeted outreach programmes.

Keywords AI adoption, AI knowledge, AI awareness, Technology acceptance, Confidence in AI, Digital literacy

1 Introduction

Artificial intelligence (AI) technologies are rapidly changing communities and societies and has had a significant impact on the way of life of individuals [7, 8]. Artificial intelligence came about around the 1950s when a group of computer scientists worked on an area called neural networks, marking the beginning of artificial intelligence [21]. John McCarthy (MIT) originally used the phrase in 1956 at a conference held at Dartmouth College. Typically, systems that exhibit intelligent behaviour through environmental analysis and conduct relatively autonomous actions to accomplish predetermined goals



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are referred to as artificial intelligence (AI) systems [21]. Many complicated tasks that were formerly completed by people may now be handled by AI, including speech recognition, picture identification, and natural language processing [25]. Artificial intelligence (AI) has become a major factor in the development of technology worldwide, with the prospect of extraordinary improvements in several fields, including healthcare, finance, agriculture and education.

The ideal situation for the adoption of AI across Ghana will be the equitable adoption of AI across diverse demographic groups, where AI is adopted to enhance productivity, facilitate education and healthcare, while mitigating risk such as job displacement and bias. However, the current adoption patterns of AI in Ghana remain unevenly spread, with several studies focusing on professions including students, academicians and health-related, while ignoring other populations proven to have adopted AI in different settings. Also, there is limited evidence on how socio-demographic factors influence the adoption of AI in non-Western settings like Ghana.

If unaddressed, this gap could leave Ghana behind in the AI revolution and limit Ghana's ability to utilise AI for national development further widening digital inequalities. This study addresses the gap by adopting the Technology Acceptance Model (TAM) in examining the knowledge of AI use and the prediction of AI adoption among residents in the Greater Kumasi area of Ghana. The objective of the study are (1) To examine how socio-demographic factors (age, education, gender) influence the perceived usefulness of AI among residents in the greater Kumasi area (2) To assess how AI knowledge and confidence shape the perceived ease of use of AI among selected residents in the Greater Kumasi area. The study links knowledge and confidence to perceived ease of use and socio-demographic factors to perceived usefulness. This framing allows this study to contribute both practically and theoretically by providing target areas for AI literacy education and extending the Technology Acceptance Model (TAM) to an under-researched context in sub-Saharan Africa. This study does not limit itself to generative AI only, it includes a diverse set of AI-powered applications that are available and accessible to the population living in the Greater Kumasi area.

2 Review of literature

Technology adoption studies, including artificial intelligence, increasingly highlight the role of demographic and contextual factors across diverse settings. In shaping uptake or technology use, understanding the influence of AI adoption across different socio-environmental contexts, especially in non-Western settings, is essential for scaling up innovations. According to Jain [18], age and educational level are significant predictors of the adoption of technology, with young and middle-aged farmers perceiving highly the usefulness and the ease of use of technology as compared to older individuals. However, gender does not appear to contribute to the adoption of technologies [18]. In Dhraief et al. [11] study, the findings revealed the importance of economic and socio-demographic factors such as education, age and income in the adoption of technologies. The application of the technology adoption model (TAM) in Jain [18] also demonstrated that several factors, including the perceived usefulness and the perceived ease of use of a technology, moderated by socio-demographic factors, could influence the adoption of artificial intelligence.

Studies on AI adoption across various sectors, though emerging, demonstrate that knowledge and awareness of AI are determinants of the adoption of AI technologies. A study which examined key issues for digital technology and AI adoption revealed that knowledge sharing and AI adoption are related to digital technologies, as knowledge was shared, the adoption of digital technologies, including AI, increased. Knowledge sharing has a favourable correlation with both privacy and security and AI adoption [4]. The awareness and adoption of AI technologies are influenced by gender, with males adopting AI more than females, whilst age and professional experience did not predict the awareness and adoption of AI [8], with AI awareness defined as having knowledge and exposure to AI. Having no knowledge or little knowledge of AI led to the adoption of AI as a starting point [23].

In the Ghanaian context, previous studies have explored AI awareness, adoption, and attitudes across different professions and academic groups. Adarkwah et al. [1] in a study among Ghanaian academics found that there was a mix of excitement and fear regarding the use of AI tools in teaching, with the majority of participants lacking a conceptual knowledge of AI tools such as ChatGPT and its potential application in learning, teaching and personal development. Contrarily, in Nyarko [19] study among Ghanaian tertiary students, reported a high level of awareness of AI (56%) and a significant degree of knowledge of AI. Edzie et al. [13] study revealed that there was awareness and adoption of AI among radiologists in Ghana. According to Ampofo, Emery and Ofori [3], nine in ten (92.3%) of medical imaging students studied had a solid understanding of AI. Despite the high knowledge and awareness of AI, Holmes et al. [16] have highlighted the importance of responsible innovation and ensuring AI aligns with human values. In the same vein, Pflanzner et al. [20] focus attention on the value alignment in AI systems, stressing that AI technologies must reflect fairness, inclusiveness, and human dignity.

Beyond demographics, behavioural factors also influence AI adoption. In their study, Hong Chuyen and Vinh [17] found that behavioural intention to utilise AI-powered design tools is not strongly correlated with facilitating conditions and trust/confidence. The non-significant findings for these two variables imply that neither the number of resources available (facilitating conditions) nor the degree of confidence and trust in the tools have a significant impact on users intention to employ AI-powered design tools. Facilitating circumstances and trust/confidence were not predictive of behavioural intention to use AI-powered design tools [17]. As new technologies continue to emerge, the adoption behaviour of these technologies by users has become crucial and a theoretical framework has emerged to help in the prediction of technology adoption [6]. The Technology Acceptance Model (TAM) is a theoretical framework for understanding and predicting user intentions to use and accept technologies developed by Davis in 1986 [10]. The TAM framework is based on two main constructs, which are perceived usefulness and perceived ease of use. Perceived usefulness has to do with an individual's intention to use a technology if that technology will enhance their performance [10]. Perceived ease of use has to do with the degree to which an individual believes that using a technology will be free of effort. TAM posits that these factors contribute to an individual's intention to use and subsequent usage behaviour of technologies [10].

Drawing on TAM constructs, this study examines whether knowledge and confidence in AI influence the perceived ease of use while socio-demographic factors like age, gender and education shape perceived usefulness. We conceptualise knowledge of AI

and confidence in understanding AI as proxies for perceived ease of use, while socio-demographics are interpreted as perceived usefulness, consistent with [11, 18]. Applying TAM to this study provides an organised way for the understanding of the adoption of AI technology among residents in the selected districts and municipalities in the greater Kumasi area. Despite there being several theories to explain technology adoption, TAM was adopted in this study due to its empirical strength in explaining individual-level behavioural intentions towards the use of technology. Unlike the diffusion of innovation theory, which emphasises macro-levels such as infrastructure, TAM constructs allow users to test how individual attitudes, rather than infrastructure, drive AI adoption by residents in the greater Kumasi area of Ghana. Dhraief et al. [11] and Jain [18] have both successfully applied TAM to technology adoption, validating its relevance, accordingly, we adopt TAM in its original form to guide the selection of variables and the interpretation of results in this study [11, 18].

Most existing studies on AI adoption focus on students, tech professionals or health-care professionals in urban centres, leaving out populations exposed to AI through everyday interactions. Additionally, the application of TAM in a non-Western context like Ghana is limited. This study fills the gap by applying TAM to examine the knowledge of AI use and the prediction of AI adoption among selected residents in the Greater Kumasi area of Ghana. Based on TAM's constructs, perceived usefulness refers to the degree to which an individual believes that using technology will enhance their performance. In this study, socio-demographic characteristics including age, gender and education are theorised to influence perceived usefulness as these factors shape how individuals perceive the relevance and utility of digital technologies [6, 18]. Perceived ease of use refers to the degree to which an individual believes that using a technology will be free of effort. These constructs reflect an individual's familiarity and comfort with engaging with AI systems [5, 15]. The conceptual framework below illustrates the theorised relationships between the study's key variables as derived from TAM.

3 Materials and methods

3.1 Settings and approach

This is a quantitative cross-sectional survey involving the use of self-administered questionnaires evaluating AI knowledge awareness, adoption and sources of exposure to AI among various socio-demographic groups among selected residents in the greater Kumasi area. The study was conducted in the Ashanti region of Ghana. Within the Ashanti region, the study's exact setting was limited to selected districts and municipalities in the greater Kumasi area, defined to include Kwadaso, Oforikrom, Ejisu, Santasi and Subin. The study focused on the greater Kumasi area, given that the region is one of the largest in Ghana with diverse socio-economic and occupational groups. The industrial nature of the region and increasing digital penetration through the emergence of AI startups make it ideal for studying AI adoption in the region.

4 Sampling and study population

Willie [24] and, Amedahe and Asamoah [2] define population as the target group the researcher is interested in gaining information and drawing conclusions. The target population is the units for which information is required and studied. The study population included all residents in the greater Kumasi area with an estimated population of

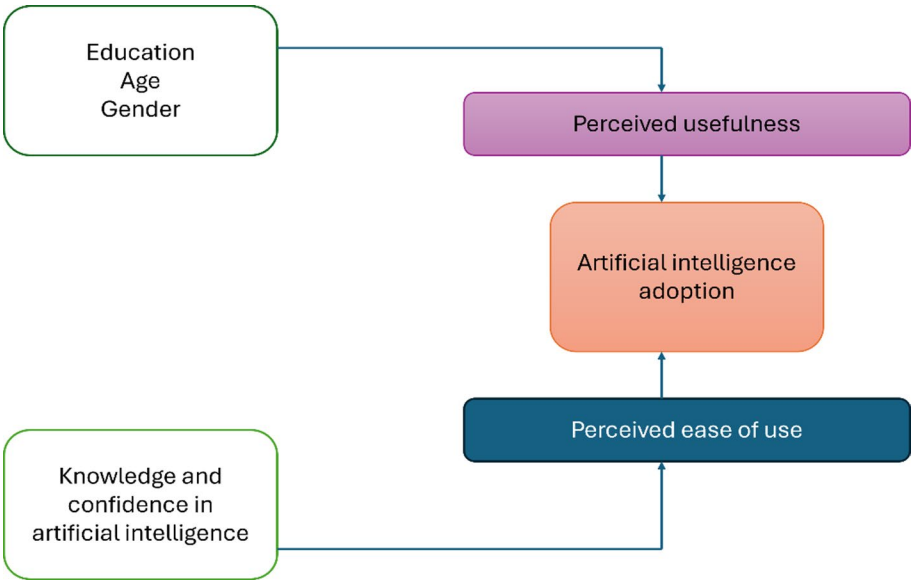


Fig. 1 Conceptual framework based on the Technology Acceptance Model (TAM)

2,627,765 [14]. A multistage sampling (cluster and purposive) technique was used for this study. Due to the absence of any known previous study in Ghana of a similar nature, the elements of appropriateness, representativeness and proportionality informed the sample size selection.

This study employed a combination of multistage sampling techniques to ensure both representativeness and accessibility of participants. The multistage sampling involved: firstly, the sampling and identification of relevant groups or subpopulations of interest based on the objectives of the study. Stage one involved the identification of the subpopulations (clusters) of interest based on their relevance to the study. These groups included university students, legal professionals/researchers, AI industry professionals and experts, traders and vendors, human rights activists and lawyers, drivers, senior high school students and junior high students. Within each cluster, a form of cluster sampling was employed by selecting specific groups, organisations or entities that are most relevant to the research objectives. Stage two characterised a purposive sampling to select participants from each cluster, a predetermined sample size was purposively selected from each cluster. This approach allowed for a comprehensive understanding by capturing perspectives from diverse backgrounds and expertise levels while maintaining a focus on the research objectives.

The Cochran formula $n_0 = Z^2 (p)(q)/e^2$ for estimating sample size for unknown population was used; where n is the sample size, z is the confidence (95%) level, e is the desired level of precision, p is the (estimated) proportion of the population which has the attribute in question and q is $1-p$ [22]. Based on the Cochran equation, a 384 sample size was obtained. The sample size was increased by 10% resulting in a sample size of 407 respondents to account for attrition. The selection of the Cochran unknown population formula is motivated by its suitability for determining the sample size of an unknown population, since the population of various clusters selected in the greater Kumasi was not known.

5 Data collection and variable measurement

Data collection commenced in September 2024 and ended in November 2024 using a structured questionnaire designed using the Kobo Collect toolbox [9]. We administered the questionnaire online through a link sent to various organisations and in person through the Kobo Collect application. Data collected was uploaded to the Kobo Collect toolbox server under password protection with limited access.

The questionnaire for this study was designed using constructs from TAM and existing validated instruments in digital literacy and self-efficacy literature. The key variables and how they were operationalised are as follows: AI adoption refers to whether an individual has used AI-powered applications [10] and is measured as the dependent variable in the binary logistic regression analysis. This approach helped us to test how the TAM constructs mediate the relationship between socio-demographic factors and adoption behaviour in the study population. Our binary scale (yes/no) used to measure knowledge of AI was adopted from the binary self-reported measure from Hargittai (2005) [15], digital literacy widely used to assess basic familiarity with technologies. Confidence in AI employed a 5-point Likert scale (1-not confident, 2-somewhat confident, 3-moderately confident, 4-very confident, 5-extremely confident) following the computer self-efficacy scale validated across multiple technology domains, including artificial intelligence [5]. Socio-demographic variables, including age, gender, and education level, were included to examine their influence on perceived usefulness in line with TAM-based studies [6, 18]. The questionnaire was pretested with a sample of 42 participants. The participants understood the questions as intended and could meaningfully differentiate between measures, making the questionnaire appropriate and clear in the context of this study.

5.1 Data analysis

The data collected was analysed using descriptive and inferential statistics aided by A STATA software version 17SE. Frequencies and percentages described sex, age, gender, religion, education level, occupation and the knowledge and awareness of AI. Inferential statistics was used to determine the association between socio-demographic factors, knowledge of AI and source of initial exposure to the adoption of AI-powered applications. The association between socio-demographic characteristics, knowledge of AI and source of initial exposure with usage of AI-powered applications was examined using a chi-square test of association, with a p-value of 0.05 considered statistically significant. The study used a binary logistic regression to analyse the effect of specific variables on AI adoption.

5.2 Ethical considerations

To uphold ethical standards, participant protection and confidentiality [12], prior ethical clearance was obtained from the Humanities and Social Sciences Research Ethics Committee (HuSSREC) with approval number (HuSSREC/AP/75/VOL.3) dated 2nd September 2024. Additionally, formal permission was secured from various institutions where the study took place and all participants provided written informed consent to participate in the study and for their data to be published. Voluntary choice to participate, without any pressure, was respected. Furthermore, withdrawal without consequences was assured. Anonymity was maintained, with unique codes used instead of personal identifiers.

6 Results

6.1 Socio-demographic characteristics of respondents

Table 1 presents the socio-demographic characteristics of respondents. Out of 407 respondents, nearly two-thirds were males, 257(63.14%). The dominant age group were those within 21–30 years, 157(38.57%) of the bracket. A majority, 342(84.03%), were Christians and had had tertiary education, 239(58.72%). Students (Junior High school, Senior High and Undergraduates) 152(37.35%) formed the dominant group studied, and the least were data scientists with 9(2.21%).

6.2 Knowledge and awareness of AI

Table 2 presents the results on the knowledge, awareness and adoption of AI use and application among study subjects, on the matter of general knowledge of AI, 342(84.03%) had knowledge of AI in terms of what it does or is used for. For knowledge of the concept of AI, eight out of every ten 338(83.05%) studied had knowledge of the concept of AI.

Social media is the dominant initial source of exposure to AI 165(40.54%). Most respondents, 320(78.62%), had experience with AI-powered applications or services. On the types of AI applications or services used, the majority, 216(47.58%) of the participants reported using AI writing assistants and the least used AI application was reported

Table 1 Socio-demographic characteristics of respondents

Variables	Frequency N (%)
Sex	
Male	257(63.14)
Female	150(36.86)
Age	
10–20	101(24.82)
21–30	157(38.57)
31–40	95(23.34)
40+	54(13.27)
Religion	
Christian	342(84.03)
Muslim	51(12.53)
Traditionalist	13(3.19)
Other	1(0.25)
Educational Level	
Junior high	83(20.39)
No formal education	21(5.16)
Secondary education	62(15.23)
Tertiary education	241(59.22)
Occupation	
AI researcher	21(5.16)
AI software developer	11(2.70)
Computer Engineer	12(2.95)
Data Scientist	9(2.21)
Driver	40(9.83)
Human right activist	16(3.93)
Law Student	46(11.30)
Lawyer	13(3.19)
Student	152(37.35)
University Researcher/Lecturer	51(12.53)
Vender/Traders	36(8.85)

Table 2 Knowledge, awareness and adoption of AI

Variables	Frequency N (%)
<i>General knowledge of AI (n = 407)</i>	
Yes	342(84.03)
No	65(15.97)
<i>Knowledge of the concept of AI (n = 407)</i>	
Yes	338(83.05)
No	69(16.95)
<i>Source of initial exposure to the concept (n = 407)</i>	
Academic sources	72(17.69)
Lecturers at school	29(7.13)
Not applicable	42(10.32)
Peers/Colleagues	82(20.14)
Social media	165(40.54)
Traditional media	17(4.18)
<i>AI application usage (n = 407)</i>	
Yes	320(78.62)
No	87(21.38)
<i>Categories of AI applications or service used (Multiple responses n = 454)</i>	
Virtual assistant	143(31.50)
Healthcare application	40(8.81)
Customer service chatbot	55(12.11)
AI writing assistant	216(47.58)
<i>What AI was used for (Multiple responses n = 511)</i>	
Set reminders, read messages	83(16.24)
Get answers to resolve issues	185(36.20)
Assess symptoms, consult remotely	32(6.26)
Generating text	140(27.40)
Language translating	71(13.90)
<i>Confidence in understanding AI (scale of 1–5, n = 407)</i>	
Not confident	82(20.15)
Somewhat confident	55(13.51)
Moderately confident	106(26.04)
Very confident	110(27.03)
Extremely confident	54(13.27)

on healthcare applications, 40(8.81%). Most participants, 185(36.20%), who used AI-powered applications or services, used them to get answers to resolve issues. On the confidence level of participants in their understanding of AI, rated on a scale of 1 to 5 (from 1-not confident to 5-extremely confident), nearly three out of every ten residents studied in the greater Kumasi area expressed being confident in their understanding of AI. several participants demonstrated being confident 110(27.03%) in their understanding of AI.

6.2.1 Association between demographic characteristics, knowledge of AI and source of initial exposure with usage of AI-powered applications

The study explores the association of AI knowledge-related variables and sociodemographic characteristics with the use and adoption of AI-powered applications by way of a chi-square test presented in Table 3. Age ($\chi^2 = 17.538$, $p = 0.001$), educational level ($\chi^2 = 105.816$, $p = 0.000$), knowledge of AI ($\chi^2 = 149.987$, $p = 0.000$) and source of initial exposure ($\chi^2 = 167.086$, $p = 0.000$) were significantly associated with the adoption of AI-powered applications.

Table 3 Association between socio-demographic factors, knowledge of AI and source of initial exposure with usage of AI-powered applications

Variables	Present use of AI applications		<i>p</i> -value	Chi-Square
	Yes-Freq (%)	No-Freq (%)		
<i>Age</i>			0.001***	17.538
10–21	76(75.25)	25(24.75)		
30–40	137(87.26)	20(12.74)		
31–40	74(77.89)	21(22.11)		
40+	33(61.11)	21(38.89)		
<i>Gender</i>			0.082	3.022
Male	209(81.32)	48(18.68)		
Female	11(74.00)	39(26.00)		
<i>Religion</i>			0.317	3.526
Christian	268(78.36)	74(21.64)		
Muslim	43(84.31)	8(15.69)		
Traditionalist	8(61.54)	5(38.46)		
Atheist	1(100)	0(0)		
<i>Educational Level</i>			0.000***	105.816
No formal education	3(14.29)	18(85.71)		
Junior high	47(56.63)	36(43.37)		
Senior education	46(74.19)	16(25.81)		
Tertiary education	224(92.95)	17(7.05)		
<i>Knowledge about AI</i>			0.000***	149.987
No	14(21.54)	51(78.46)		
Yes	306(89.47)	36(10.53)		
<i>Source of initial exposure</i>			0.000***	167.086
Academic source	61(84.72)	11(15.28)		
Lecturers at school	21(72.41)	8(27.59)		
N/A	1(2.38)	40(97.62)		
Peers/Colleagues	74(90.24)	8(9.76)		
Social media	148(89.70)	17(10.30)		
Traditional media	15(88.24)	2(11.76)		

*** $p < 0.05$

6.3 Predictors of AI adoption

The independent predictors of the adoption of AI-powered applications were explored using a binary logistic regression presented in Table 4. In the unadjusted crude model, the age of participants is a significant predictor of the adoption of AI-powered applications. Respondents aged 21–30 [OR 2.253 (95.0% CI (1.174–4.322; $p = 0.015$))] had significantly higher odds of adopting AI-powered applications compared to those within the age ranges of 10–20.. Respondents at the junior high school level [OR 7.833 (95.0% CI 2.141–28.658; $p = 0.002$)], secondary education [OR 17.25 (95.0% CI 4.480–66.417; $p = 0.000$)] and tertiary education [OR 79.058 (95.0% CI 21.162–295.347; $p = 0.000$)] are significantly more likely to have adopted AI compared to those with no formal education. In relation to knowledge of AI, the model revealed that participants with knowledge of AI are 30.964 times more likely to adopt AI-powered applications as compared with those without knowledge [OR 30.964 (95.0% CI 15.611–61.414; $p = 0.000$)]. Persons' who are somewhat confident [OR 20.1 (95.0% CI: 8.296–48.694; $p = 0.000$)], moderately confident [OR 74.444 (95.0% CI: 27.495–201.560; $p = 0.000$)], very confident [OR 159.311 (95.0% CI 44.443–571.065; $p = 0.000$)] and extremely confident [OR 236.733 (95.0% CI 30.2918–1850.094; $p = 0.000$)] have significantly higher odds of adopting AI-powered applications as compared to those who are not confident in AI-powered

Table 4 Binary logistic regression on the independent predictors of AI adoption

Variables	OR(CI)	P-value	AOR(CI)	P-value
<i>Age</i>				
10–20(ref)	1			
21–30	2.253(1.174–4.322)	0.015***	0.504(0.153–1.662)	0.261
31–40	1.159(0.597–2.248)	0.662	0.904(0.235–3.477)	0.884
40+	0.516(0.254–1.050)	0.068	0.565(0.133–2.397)	0.439
<i>Gender</i>				
Female(ref)	1			
Male	1.529(0.945–2.475)	0.083		
<i>Educational level</i>				
No formal education(ref)	1			
Junior high	7.833(2.141–28.658)	0.002***	0.788(0.098–6.328)	0.823
Secondary education	17.25(4.480–66.417)	0.000***	0.675(0.075–6.053)	0.726
Tertiary education	79.058(21.162–295.347)	0.000***	4.968(0.691–35.675)	0.111
<i>Religion</i>				
Others(ref)	1			
Muslim	2.986(0.791–11.272)	0.107		
Christian	2.012(0.654–6.185)	0.222		
<i>Knowledge of AI</i>				
No(ref)	1			
Yes	30.964(15.611–61.414)	0.000***	3.813(1.340–10.850)	0.012 ***
<i>Confidence in AI</i>				
Not confident(ref)	1			
Somewhat confident	20.1(8.296–48.694)	0.000***	12.337(4.074–37.356)	0.000***
Moderately confident	74.444(27.495–201.560)	0.000***	41.616(12.911–134.136)	0.000***
Very confident	159.311(44.443–571.065)	0.000***	64.270(15.317–269.667)	0.000***
Extremely confident	236.733(30.2918–1850.094)	0.000***	82.843(9.320–736.332)	0.000***

***: $p < 0.05$

applications. Gender did not show a significant association with the adoption of AI-powered applications.

In the adjusted model, controlling for other variables, confidence in AI and knowledge significantly predicted whether a person adopts AI application platforms or not. When adjusted for cofounders, in the multilevel regression model, knowledge of AI remained a significant predictor of AI adoption [Yes (AOR 3.813 (95.0% CI 1.340–10.850; $p = 0.012$)). Additionally, confidence in AI is a strong predictor of adoption of AI: participants who are somewhat confident [AOR 12.337 (95.0% CI 4.074–37.356; $p = 0.000$)], moderately confident [AOR 41.616 (95.0% CI 12.911–134.136; $p = 0.000$)], very confident [AOR 64.270 (95.0% CI 15.317–269.667; $p = 0.000$)] and extremely confident [AOR 82.843 (95.0% CI 9.320–736.332; $p = 0.000$)] are significantly more likely to have adopted AI-powered applications compared to those who are not confident.

7 Discussion

This study investigated the knowledge of AI use and the prediction of AI adoption in selected municipalities and districts within the greater Kumasi area of Ghana. In relation to knowledge of AI, in all, eight in ten of residents in the selected municipalities and districts know of and are aware of AI. The initial exposure to AI is mostly through social media and most residents (78.62%) in the greater Kumasi area adopted and used AI-powered applications or services. Mostly, residents in the greater Kumasi area use AI-powered applications or services to get answers to resolve issues and most of them

were confident in their understanding of AI. Knowledge of AI and confidence in understanding AI can lead to adopting and using AI-powered applications or services. These findings suggest how knowledge is a powerful and driving tool in the adoption of AI-powered applications, as awareness plays a crucial role in informing and familiarising users with technology, AI education must be done especially for populations that have limited exposure to formal education channels to provide AI knowledge which may lead to the adoption of AI.

Age and educational level were seen to be associated with the adoption of AI. The influence of age and educational level as associated with the adoption of AI technologies have been discussed in the literature. For instance, Dhraief et al. [11] in their study found that demographic characteristics such as age, gender, family income and size, educational qualification and beliefs have a significant positive relation to the adoption of technologies. Additionally, knowledge of AI and the source of initial exposure were both found to be significantly associated with the adoption of AI-powered applications. This might be because as one grows and continues to engage in education, their educational level, knowledge and age increase, which explains why these factors are significant in adopting AI.

These findings of high knowledge and awareness of AI are consistent with earlier studies by [13, 19]. In their study among radiologists, Edzie et al. [13] found a high level of awareness (44.2%) of AI among radiologists. Nyarko [19] work among Ghanaian tertiary students also found a high level of awareness of AI (56%) and a significant degree of AI knowledge. A similar pattern was confirmed by Ampofo, Emery and Ofori [3] in their study among medical imaging students revealed that (92.3%) of participants had a solid understanding of the concept of AI. This study is, however, unique in that it emphasises on how informal learning channels such as social media play a key role in AI education. This suggests that a focus on nontraditional platforms in AI education when exploring technological literacy should be considered. Notwithstanding, the findings contradict Adarkwah et al. [1] study among Ghanaian academics, where the majority Ghanaian academics lacked knowledge of ChatGPT. The divergence between earlier study findings and Adarkwah et al. [1] suggests that over-reliance on formal educational systems to promote AI knowledge may be insufficient and informal sectors such as social media may provide alternative means to promoting awareness and knowledge of AI. A time-lag difference can also be attributed to the differences between this study's findings and Adarkwa et al. [1]. Findings in this study directly address the gap in prior literature, where studies in Ghana, although limited, focused on professions including healthcare professionals, students and academicians. By including a broader demographic sample, our study expands the understanding of AI adoption in diverse digital contexts. Beyond academics and to the larger populace, the present study demonstrates how informal learning channels such as social media can be critical in AI education, contributing to the literature by showing how TAM's constructs operate outside of Western or formalised digital environments and remain validated across different population groups.

Our regression model revealed that both age and education significantly predicted AI adoption and use, though a person's gender did not appear to have any effect on whether a person would use or adopt AI. Age might have been significant because respondents aged 21–30 fall into the millennial and early generation Z category, who have grown up in an era of rapid technological transformation, making them more accustomed to and

open to integrating technology into their daily lives. Education might have been significant because as people progressed in their education level, the likelihood of AI use increases sharply since they must have seen its benefits and also developed the capacity to use it. Gender has no significant association with the adoption of AI-powered applications. Gender might not have been a predictor of AI adoption because key determinants such as age and educational level had a stronger influence on it. These factors may have overshadowed gender-based differences, aligning with trends indicating a narrowing gender gap in technology use as access and familiarity with technology grow. Our findings are consistent with previous studies that report age and education to be significant factors in the adoption of technology, with young and middle-aged individuals perceiving the usefulness of technology and the ease of use of technology as compared to older individuals [18]. Gender was not seen as a significant factor in contributing to the adoption of technologies [18]. However, this contradicts D'Souza [8], whose study revealed that the awareness and adoption of AI technologies were influenced by gender, with males adopting AI more than females, whilst age and professional experience did not influence the awareness and adoption of AI [8].

In relation to knowledge of AI, the study results reveal that knowledge of AI is a significant predictor of AI adoption. Similarly, in a study which examined knowledge sharing key issues for digital technology and artificial intelligence adoption, results revealed that knowledge and AI adoption are positively correlated with digital technologies. Knowledge sharing has a favourable correlation with both privacy and security and AI adoption [4]. Our findings rather deviate from earlier observations made by Tully, Longoni and Appel [23], where no knowledge of AI or lower levels of AI knowledge predicted the adoption of AI, since such individuals were surprised by the capabilities of AI. Further to this study's findings, confidence in AI significantly predicts the adoption of AI-powered applications. However, previous works by Hong Chuyen and Vinh [17], where facilitating circumstances and trust/confidence were not predictive of behavioural intention to use AI-powered design tools for several reasons, emerge as contradictory findings to our present results [17]. The study set out to achieve two objectives (1) to examine how socio-demographic factors (age, education, gender) influence the perceived usefulness of AI among residents in the greater Kumasi area, and (2) to assess how AI knowledge and confidence shape the perceived ease of use of AI among selected residents in the Greater Kumasi area. The findings show that education and age (21–30 years) was a significant predictor of perceived usefulness of AI, while AI knowledge and confidence strongly influenced perceived ease of use and adoption. This outcome aligns with the TAM and confirms that some demographic factors and psychological factors play critical roles in shaping AI-related behaviour.

In our multilevel analysis, knowledge of AI remains a significant predictor of AI adoption, together with a person's confidence in AI, consistent with the constructs of the Technology Acceptance Model (TAM) framework. Factors, such as education and age, influence the perceived usefulness and ease of use of AI technologies [18], and as such with higher levels of education or younger ages are likely to perceive AI as more beneficial, reflecting TAM's construct of perceived usefulness. The socialisation of AI use is also constructed with generation preferences, such as the millennial generation, who are within their acceptance mid-twenties and late twenties, are comfortable with high-performing technology solutions and remain technology savvy.

More importantly, confidence in understanding AI (reflecting perceived ease of use) significantly correlates with adoption, indicating that greater familiarity and comfort with AI technology reduces perceived barriers to its usage. This application of TAM highlights how socio-demographic variations impact AI adoption through personal perceptions, especially in the context of non-Western societies where technological diffusion is emerging [11]. Our results share in similar pattern, validating the utility of TAM in technology uptake and innovation diffusion, particularly in the case of AI adoption for the Ghanaian community.

The study contributes to the larger sociology of technology field and more specifically to the application to TAM in a non-Western context, particularly among some selected residents in the Greater Kumasi area of Ghana. For policymakers, the results highlight the need for inclusive digital education policies that prevent some populations from being left behind in the era of AI deployment. For practitioners, this study suggests targeted AI programmes to build confidence and understanding of AI, particularly for older adults and those with limited formal education. For society, the study offers useful insights into the demographic spectrum that influences who benefits from emerging technologies. The use of non-traditional avenues for facilitating such technology uptake has proven to be useful in this study.

8 Conclusion

This study contributes to the understanding of AI knowledge and adoption by analysing the phenomenon that influences knowledge, awareness and adoption of AI-powered applications. The study findings demonstrate that the level of education, age, knowledge and confidence in AI are determinants of AI adoption, which reflects the literature on AI, reflecting characteristics that are consistent with technology acceptance literature. By applying the Technology Acceptance Model (TAM), this study illustrates how perceived ease of use and usefulness, shaped by socio-demographic factors, affect AI adoption. This study significantly contributes to the literature on AI adoption in non-Western settings. Particularly in sub-Saharan Africa, given the limited research in this area. The results offer crucial insights into AI diffusing its pathways in Ghana, which can be a starting point for policymakers to leverage the results of this study to develop tailored AI education programmes that address socio-demographic disparities and promote a more inclusive AI design and deployment process. Future research is required to investigate the influence of additional factors such as digital infrastructure and cultural issues in the adoption of AI in the context of developing countries.

8.1 Limitations and strengths

Our cross-sectional survey design is limited by its inability to determine causality between socio-demographic characteristics and AI adoption. Additionally, the study adopted a monomethod without methodological triangulation, which may constrain the depth of insight into underlying social dynamics. The use of purposive sampling and the study's focus on some selected districts and municipalities in the greater Kumasi area may also pose selection biases, which could affect the generalisation of the results. Additionally, confidence in AI might not uniformly translate into accurate knowledge of AI since self-reported biases may compound social desirability bias. By focusing on socio-demographic factors in the greater Kumasi area, the study contributes and fills the gap in

AI adoption literature, particularly in non-Western settings. The study's robust methods and statistical methods ensure the reliability of the study's results.

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Author contributions

E.S. led the study and was primarily responsible for writing the manuscript. S.C.A. contributed to the conceptualization of the study and critically reviewed the manuscript at all stages. C.A. and E.A. contributed to the literature and reviewed the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets analyzed during the current study are not publicly available due to confidentiality agreements or privacy concerns but are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare no competing interests.

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